Laptop Price Prediction

Using Machine Learning

Predicting laptop prices from real-world multi-source data using regression

Samarth Singh

Aspiring Data Analyst

The Problem

Laptop prices vary dramatically across brands, specs, and regions — yet buyers and businesses often lack a systematic way to estimate fair value.

Before

- Prices compared manually — time-consuming
- No benchmark for fair market value
- Buyers overpay for equivalent specs

After

- Instant price estimate from specs alone
- Data-driven benchmark for any configuration
- Transparent, explainable predictions

Data Collection

Four real-world Kaggle datasets merged after rejecting a 1.2M-row synthetic dataset detected during validation.

Source	Key Features	Rows	Currency	Status
ds01 — laptop_price01	Brand, Processor, RAM, Storage, GPU, OS	885	INR	✓ Used
ds02 — laptop_price02	Brand, Processor, RAM, Storage, GPU, OS, Price	886	INR	✓ Used
ds03 — laptop_price03	Company, CPU, Memory, GPU, OpSys, Weight	1,270	EUR→INR	✓ Used
ds04 — laptop_price04	Brand, Processor, RAM, Storage, GPU, OS	1,013	INR	✓ Used

Data Cleaning Pipeline

Each dataset cleaned independently before merging — ETL approach: Extract → Transform → Load.

Per-Source Cleaning

- Fixed encoding artifacts (Unicode â, superscript ᵉ)
- Extracted numeric RAM/Storage from strings ('8GB DDR4' → 8)
- Standardized OS into 8 clean categories

Data Quality Fixes

- Dropped 23 corrupt rows (data-shift in storage/OS columns)
- Corrected DDR6 → DDR5 (non-existent standard)
- Fixed 1000 GB → 1024 GB TB notation

Schema Merge

- Aligned 4 different schemas to 9 common columns
- EUR → INR currency conversion (ds03)
- Removed 135 exact duplicates post-concat

Master Dataset — Final State

3,918
Total Rows

10
Columns

4
Sources Merged

135
Duplicates Removed

0
Null Values

Final Schema

brand	Categorical	Laptop manufacturer
processor	Text → extracted	Raw CPU string
gpu	Text → extracted	Raw GPU string
ram_gb	Numeric (int)	RAM in GB
ram_type	Categorical	DDR4 / DDR5 / LPDDR5 / Unknown
storage_gb	Numeric (int)	Storage in GB
storage_type	Categorical	SSD / HDD / eMMC
os	Categorical	Operating system
price	Numeric (INR)	Target variable

Price Range

₹8,000
Minimum

₹6,64,425
Maximum

Median: ₹74,390

EDA — Key Findings

01

Price is Right-Skewed

Skewness = 2.08. Most laptops cluster in ₹30K–₹1.5L, with a long premium tail. Log-transform required before modeling.

02

RAM is the Strongest Predictor

Correlation r = 0.56 with price. Every RAM tier step (8→16→32 GB) shows a consistent, monotonic price jump.

03

Brand Drives Significant Premium

Razer, Apple, and Microsoft command 2–4× the median price of budget brands like AXL or Primebook.

04

Storage Type Matters More Than Size

SSD vs HDD creates a larger price gap than doubling storage capacity — type is a stronger signal than quantity.

05

OS and Brand Are Correlated

macOS = Apple exclusively. Including both causes multicollinearity — addressed in feature engineering.

06

Outliers Are Real Premium Products

IQR outliers are Razer/Apple workstations, not data errors. Kept all; log-transform handles their leverage.

Feature Engineering

Transformed 10 raw columns into 44 model-ready features.

cpu_brand
from: processor string
→ Intel / AMD / Apple / Qualcomm / MediaTek

cpu_tier
from: processor string
→ i3 / i5 / i7 / i9 / Ryzen 3–9 / Apple Silicon

gpu_type
from: gpu string
→ Dedicated (NVIDIA/AMD) vs Integrated

brand_encoded
from: brand (target encoding)
→ Median log-price per brand → 1 dense numeric column

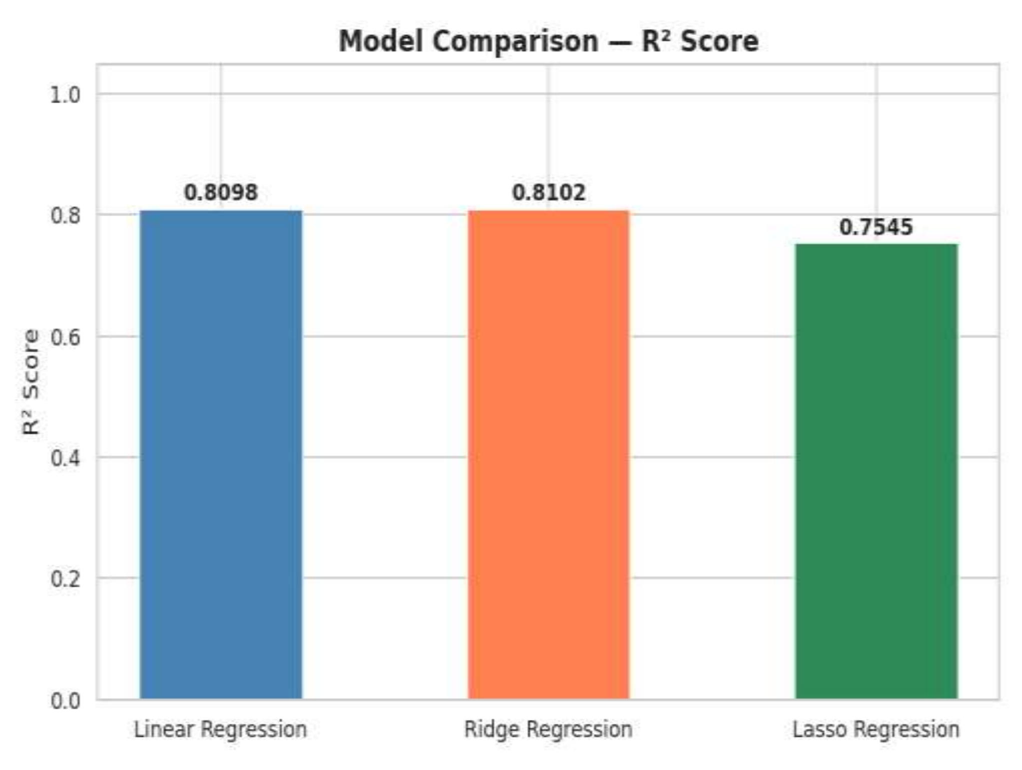
price_log
from: price
→ log1p(price) — reduces skewness from 2.08 → 0.04

Encoding Strategy

brand	Target encoding
os	One-hot (8 cats)
storage_type	One-hot (3 cats)
ram_type	One-hot (6 cats)
gpu_type	One-hot (2 cats)
cpu_brand	One-hot (6 cats)
cpu_tier	One-hot (11 cats)

44 total features after encoding

Model Results



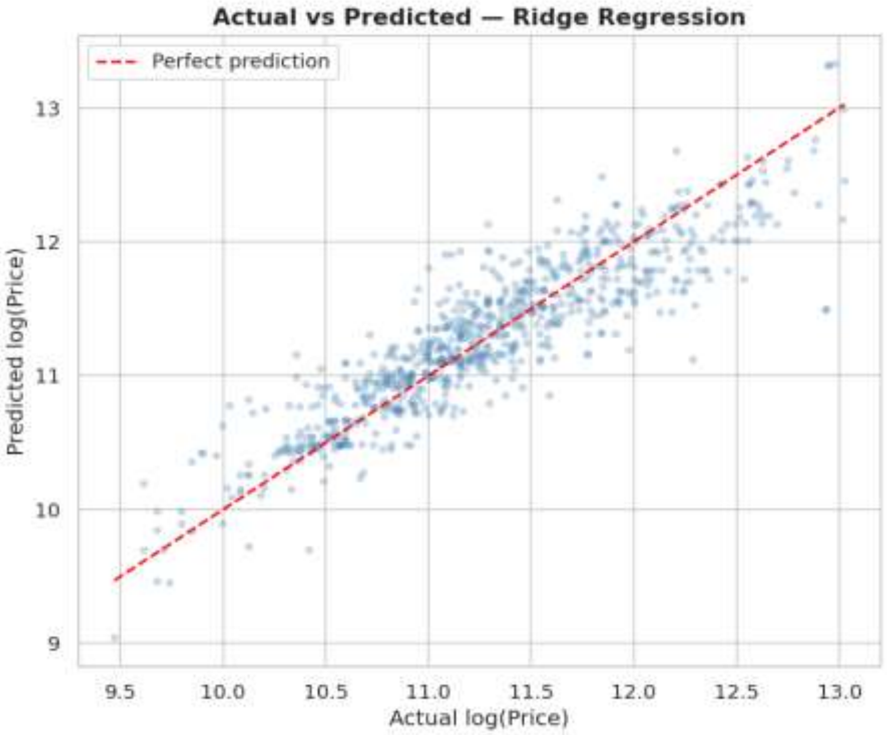
Model	R^2	MAE (INR)
Linear	0.8098	₹23,557
Ridge ★	**0.8102**	**₹23,560**
Lasso	0.7545	₹27,069

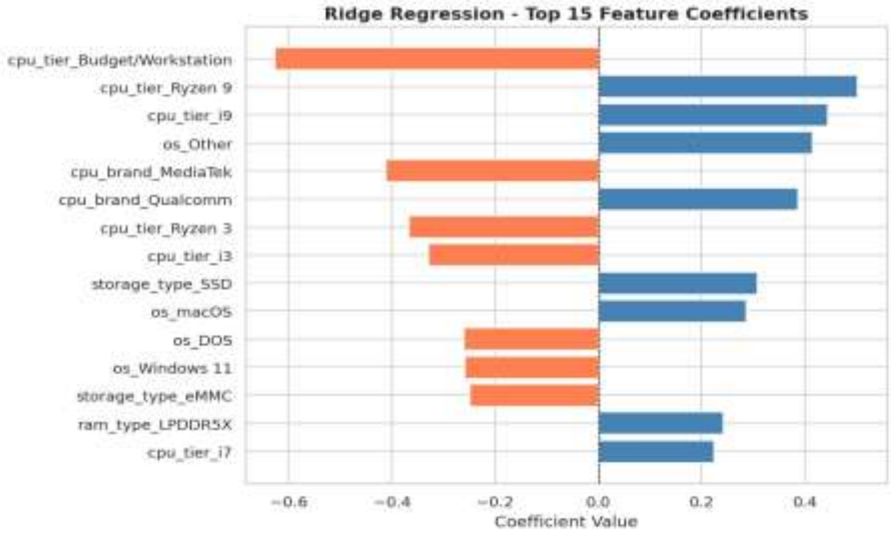
Ridge is the best model
R^2 = 0.8102 — explains 81% of price variance on test set.

MAE ≈ ₹23,500 in practice
Average prediction error on a ₹74K median price laptop.

Lasso over-eliminated features
L1 penalty removed columns still contributing real signal.

Ridge Regression — Deep Dive





- Points hug the diagonal → strong predictive accuracy across all price tiers
- cpu_tier_Ryzen 9 & i9 = strongest positive price drivers
- cpu_tier_Budget/Workstation = largest negative coefficient (price penalty)
- SSD storage type adds consistent positive premium over HDD

Key Takeaways

Data Engineering is the Real Work

5 datasets evaluated, 4 merged, 1 rejected after systematic validation. ETL pipeline built per-source before combining.

Feature Extraction > Raw Encoding

Reduced 400+ unique processor/GPU strings to 3 clean columns. Target encoding for brand avoided 36 sparse dummies.

Ridge Regression Wins — $R^2 = 0.81$

81% of price variance explained on a real-world merged dataset — without display size, battery, or weight data.

Log-Transform is Non-Negotiable

Price skewness of 2.08 would have severely biased linear models. log1p reduced skewness to near-zero.

Thank You

Laptop Price Prediction Using ML

github.com/OfficialSamarthsingh

linkedin.com/in/samarth-singh-official

samarthsinghportfolio.netlify.app

Open to Data Analyst internship and entry-level opportunities